O-NYX / SentinelVault: Project Timeline Table

One medium-risk file followed from arrival at a gateway (T+0) to disposal, intelligence sharing and twin rebuild. Timestamps are nominal design targets (T+ seconds from file arrival), to be replaced by measured values from the prototype. Low-score files pass in about 1 to 21 s; high-score files are blocked at about T+21 s with no twin created.

Step	Timestamp	Duration	Component / step	What happens: checks, processes and parameters
PHASE A | PRE-EXPLOITATION (T+0 s to T+21.3 s): triage, produces a risk score only				
A1	T+0.0 s to 0.1 s	0.1 s	**Gateway capture (Stage 0)**	Email gateway/MTA, web proxy, DNS resolver, CASB, internal file-share filter or USB control intercepts the file and holds it (email stays queued, download stays pending). A copy is forwarded to the central AI pre-screening engine. **Parameters:** attachment, embedded links, sender identity, file hash, source gateway. No inspection happens on the employee's device; USB control is the only per-endpoint exception.
A2	T+0.1 s to 1.1 s	about 1 s	**Layer 1: fast checks (run in parallel)**	**Hash reputation lookup:** MD5/SHA-1/SHA-256 against local blocklist and threat-intel feeds (hit/miss + confidence). **YARA rules:** packer signatures, malware-family byte patterns, suspicious strings, obfuscation markers (rules matched + severity). **ML static classifier:** file entropy, PE header anomalies, import-table API calls (VirtualAlloc, WriteProcessMemory, CreateRemoteThread), sections, packer detection, embedded URLs/IPs/keywords (malicious probability 0-1). **Phishing-URL check:** domain age, SSL validity, homoglyph/typosquat detection, URL reputation, redirect chain (phishing score). Raw signals go forward; Layer 1 makes no pass/block decision.
A3	T+1.1 s to 21.1 s	about 20 s	**Layer 2: deep checks (run in parallel)**	**Deep static analysis:** control-flow anomalies, embedded PE/script objects, VBA macros, packing ratio, per-section entropy, digital-signature validity, compiler timestamp anomalies. **CDR:** removes macros, OLE objects, embedded JavaScript and executables and rebuilds a safe copy (runs regardless of verdict). **Honeypot network observer:** decoy credential use, decoy file access, canary-token triggers, lateral-movement probes, honeytoken DNS requests. **Ransomware time machine:** continuous snapshots, baseline vs abnormal I/O, entropy change on file writes. Timeout 30 s per check.
A4	T+21.1 s to 21.2 s	0.1 s	**Layer 3: attacker-intent engine (score fusion)**	Fuses all Layer 1 and Layer 2 signals (hash result, YARA count/severity, ML probability, phishing score, static report, CDR findings, honeypot signal, time-machine flag) with reliability weights (a hash match outweighs one YARA hit). **Output:** one 0-100 trust/risk score. Bands: 0-30 low, 31-70 medium, 71-100 high. This is a risk rating, not a malware-type label.
A5	T+21.2 s to 21.3 s	0.1 s	**Layer 4: policy / decision engine**	Maps the score to an action using base thresholds, then organisation policy (for example stricter limits for finance or HR) and the human-override hook. **Low (0-30):** pass, no delay. **Medium (31-70):** sandbox, file shows as pending. **High (71-100):** block, quarantine, SOC alert, no twin created. Only a medium score continues to Phase B.
PHASE B | MID-EXPLOITATION (T+21.3 s to T+381.3 s): behavioural confirmation in a disposable digital twin				
B1	T+21.3 s (plus queue wait)	0 s if a slot is free	**Slot acquisition on the shared analysis cluster**	Orchestrator looks for a free slot on the shared hypervisor cluster (isolated security zone). If all slots are busy the file stays held at the gateway until one frees, so nothing reaches an inbox before its verdict. **Parameters:** number of slots, current occupancy, queue length.
B2	T+21.3 s to 61.3 s	about 40 s	**Digital-twin instantiation**	Clones stored golden images into a fresh isolated virtual network: fake domain controller, file server, email server, database, backup server, 2 to 5 representative employee endpoints, and the detonation-point VM. One twin blueprint per organisation, one disposable instance per detonation. Built from roles, not headcount; the rest of the organisation is synthetic directory data.
B3	T+61.3 s to 66.3 s	about 5 s	**Covert agent attach and realism check**	Hypervisor-level AI agent attaches to every twin VM using Virtual Machine Introspection, below the guest OS so nothing inside can see or disable it. **Channels started:** memory introspection, syscalls, process creation, network traffic inside the twin, filesystem and registry changes. Realism check compares twin hostnames, domain structure and folder conventions with the real network.
B4	T+66.3 s to 366.3 s	up to 300 s	**Detonation and full-run observation**	File is placed on the detonation VM and runs as it would on a real laptop. Agent runs four functions: **behaviour watcher** (logs every event), **intent classifier** (groups events into patterns, e.g. mass renaming + extension change), **ATT&CK mapper** (e.g. T1003, T1486, T1071), **risk scorer** (running behavioural score). Agent stays silent until the run finishes or the timeout is hit, then writes one batch report.
B5	T+366.3 s to 376.3 s	about 10 s	**Threat classification and verdict**	First point where the threat type is decided: ransomware, trojan, worm, RAT, botnet client or benign, from the observed behaviour. If behaviour matches no known pattern it is raised as a zero-day candidate (human review required). The gateway then releases or blocks the held file, so the employee verdict arrives about 6.3 minutes after arrival.

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B6	T+376.3 s to 381.3 s	about 5 s	**Blast-radius analysis and containment recommendation**	Uses the twin's network topology to work out which real assets the sample would reach first, second and third. Prepares a recommended response (isolate X, block Y, restore Z from backup). Never auto-executed: a human must approve any containment action.
PHASE C | POST-EXPLOITATION, TRACE (T+366.3 s to T+438.3 s): evidence, disposal, intelligence, rebuild				
C1	T+366.3 s to 381.3 s	about 15 s	**Evidence harvest**	Collected at end of run, before teardown: **memory dump** of the detonation VM, **network capture (PCAP)** of every packet in the twin, **filesystem diff** (created, modified, deleted files), and the **covert agent's full report** (behaviour log, intent classification, ATT&CK mapping).
C2	T+381.3 s to 383.3 s	about 2 s	**Hash-chaining, signing and sealing**	Each evidence item is cryptographically linked to the previous one (tampering with any piece breaks the chain), the bundle is signed, and it is committed to the immutable audit store. Disposal is not allowed to start until this step completes.
C3	T+383.3 s to 398.3 s	about 15 s	**Verified disposal**	**VM termination** (every twin VM shut down), **memory zeroing**, **disk cryptoshredding** (encryption keys destroyed), **VPC teardown** (virtual network deleted). A signed disposal certificate proves the teardown was complete. Runs after every detonation, malicious or not. The analysis slot is released at about T+398 s.
C4	T+383.3 s to 393.3 s	about 10 s (parallel with C3)	**Intelligence extraction and distribution**	**IOC package** (hashes, IPs, domains, file paths) converted to STIX/TAXII and pushed to MISP. **ATT&CK technique map**. **TTP-similarity report** against known campaigns (behaviour only, no attribution). **Zero-day candidate flag** only if no known pattern matches, human-confirmed; about 90-day embargo before public sharing.
C5	T+398.3 s to 438.3 s (then periodic)	about 40 s, background	**Twin rebuild and learning**	A fresh twin instance is pre-built for the next detonation. Three things persist across every rebuild: the **immutable audit store**, the **threat-intelligence database** and the **retrained classifier**. Everything else is torn down and rebuilt clean. Retraining runs in batches, not per file.